# Lecture 4

**Ananya Roy**
Lecturer
Environmental Science and Engineering
Jatiya Kabi Kazi Nazrul Islam University

# *Determination of Liquid Limit (LL) of Soil*

- Liquid limit determined in lab using Casagrande's standard liquid limit apparatus

Required apparatus includes:

- Casagrande apparatus
- Sieve No. 40 (ASTM)
- Spatula
- Containers
- Tray
- Electric oven
- Balance





# *Testing Procedure of LL*

**1. Sample Preparation**

- Take 120 g of air-dry soil passing 425 µm (No. 40 sieve).
- Add distilled water gradually and mix thoroughly until the soil becomes a smooth, uniform paste.

**2.Preparing Soil in the Cup**

- Place a portion of the paste into the Casagrande cup.
- Spread it with the spatula to form a layer about 10 mm (1cm) thick at the center and 8 mm at the edges.
- Use the grooving tool to cut a clean, sharp groove along the center of the soil pat.



*Figure 5.2: Preparation of soil slurry*



*Figure 5.3: Placing the soil paste on the Casagrande apparatus*



*Figure 5.5: The groove at the middle of the soil sample before the application of the blows*



*Figure 5.6: The groove at the middle of the soil sample after the application of the blows*

# *Testing Procedure of LL*

3. **Operating the Device**

- Turn the handle at 2 revolutions per second.
- Count the number of blows required for the soil groove to close along a length of 10-12 mm. No of blows should be between 15-35.

**If the soil slides instead of flowing, discard the result and repeat the test until proper flow occurs.**

- Collect about 10 grams of soil near the closed groove for water content measurement. (Water content is measures using balance, subtracting the weight of soil sample taken from the apparatus and weight after 24 hrs oven dry soil)
- Need 4 readings for blows around 15–35 (e.g., 35 blows, 28 blows, 22 blows, 15 blows)

Determine the **water content** by the oven-dry method:

- Weigh **Wet soil (Ww)**
- Oven dry at 105–110°C for 24 hrs
- Weigh **Dry soil (Wd)**
- Water content, $w = \frac{W_w - W_d}{W_d} \times 100$

## LIQUID LIMIT TEST

**Sample Data Sheet**

| Trial no. | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| No. of blows | 15 | 21 | 25 | 31 | 35 |
| Wt. of container in gm. | 7.7 | 11.3 | 11.1 | 7.0 | 7.3 |
| Wt. of container + wet soil, gm | 27.6 | 28.3 | 31.7 | 26.7 | 26.6 |
| Wt. of container + dry soil, gm | 19.2 | 21.5 | 23.7 | 19.2 | 19.4 |
| Wt. of water, $W_w$ in gm. | 8.4 | 6.8 | 8.0 | 7.5 | 7.2 |
| Wt. of dry soil, $W_s$ in gm. | 11.5 | 10.2 | 12.6 | 12.2 | 12.1 |
| Water content, w in % | 73.0 | 66.7 | 63.5 | 61.5 | 59.5 |

$W_{water}$

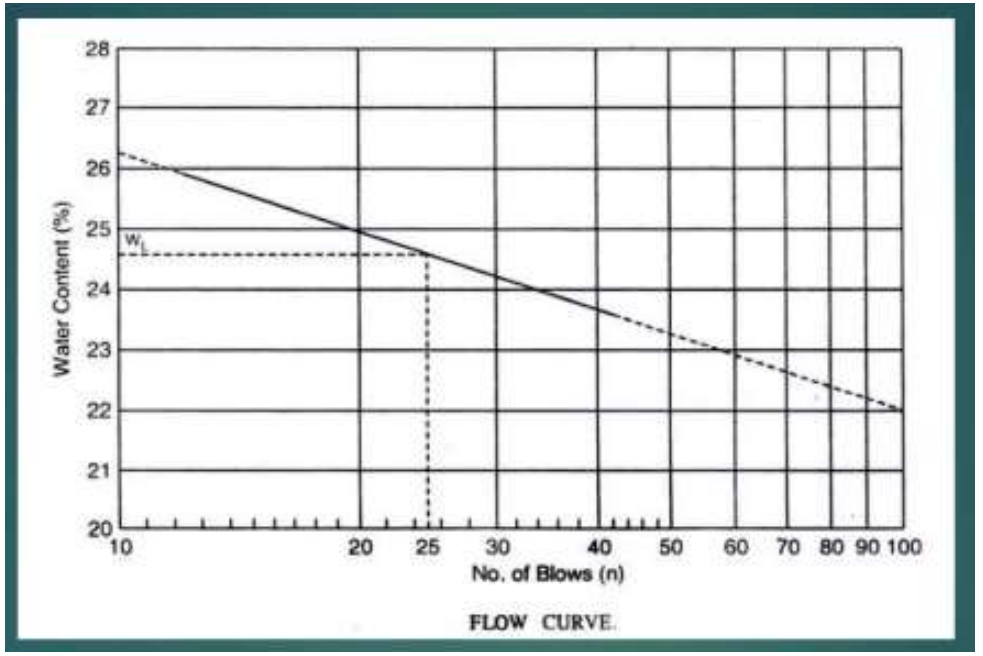
# *Testing Procedure of LL*

5. **Plotting the Flow Curve**

- Plot water content (w) on the Y-axis and log of number of blows (N) on the X-axis.
- Draw a best-fit straight line (Flow Curve).

**6. Determining Liquid Limit**

- Read the water content corresponding to N = 25 blows from the flow curve.
- This water content is the Liquid Limit (LL) of the soil.



# *Determination of Plastic Limit (PL) of Soil*

**Apparatus Needed:**

- Glass plate (2 x 2 feet)
- Spatula
- No. 40 sieve (ASTM standard)
- Electric oven
- Balance

# *Procedure of testing Plastic Limit (PL)*

**1. Sample Preparation**

- Take 20 g of air-dry soil passing 425 µm (No. 40 sieve).
- Add a little water and mix thoroughly until the soil becomes a plastic, workable mass—not too sticky, not too dry.

**2. Rolling the Thread**

- Take about 10 g of the prepared soil.
- Place it on the glass plate.
- Roll it back and forth using your fingers until it forms a thread of 3 mm (⅛ inch) diameter.

# *Procedure of testing Plastic Limit (PL)*

**3. Observe Crumbling**

- If the thread cracks or crumbles before reaching 3 mm thickness it means Soil is too dry.
- Add a little water and repeat.
- If the thread can be rolled below 3 mm without cracking that means Soil is too wet.
- Dry it slightly and repeat.
- The correct moisture condition is when the soil just crumbles at exactly 3 mm diameter.

**4. Then determine the moisture content as previous**.

Repeat rolling at least 2–3 times, and take the average water content as the Plastic Limit.



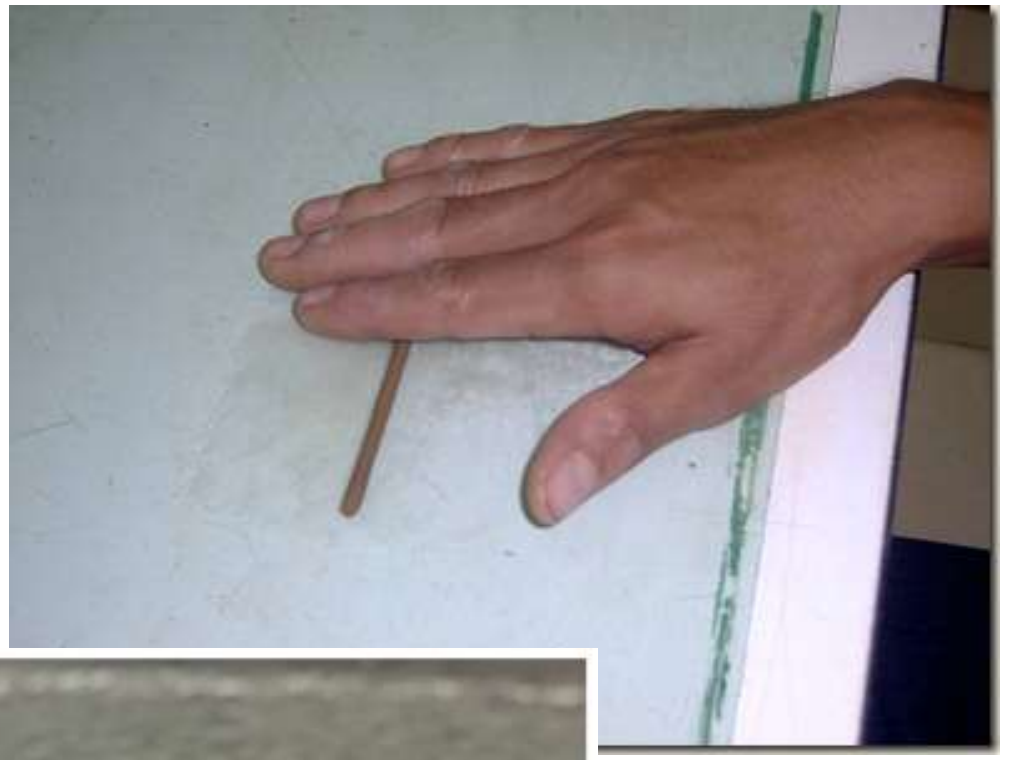


Figure 5.7: Sample preparation for the plastic limit test



Figure 5.8: A sample thread of 3mm in diameter

# *The Plasticity Index (PI)*

**The Plasticity Index (PI)** is the range of water content over which a soil remains plastic.

$$PI = \text{Liquid Limit (LL)} - \text{Plastic Limit (PL)}$$

It represents how much water the soil can absorb while still remaining deformable without cracking.

# *Check the link*

- https://www.youtube.com/watch?v=nogD7maQMJs
- https://www.youtube.com/watch?v=b9HvDB8G90k
- https://www.youtube.com/watch?v=4HNGXwqX4L0

# *Shrinkage limit test*

- It determine **the moisture content** at which further drying of soil does NOT cause any more volume reduction.

**Equipment Required**

- Shrinkage dish (45 mm diameter, 15 mm height)
- Mercury or wax-coated dish (for volume measurement)
- Glass plate
- Spatula
- Balance (0.01 g accuracy)
- Oven
- Desiccator
- Mercury tray (if using mercury method)
- Water



# *Procedure of Testing the Shrinkage limit*

**1. Preparation of Soil Paste**

- Take about 30 g of soil passing 425 µm sieve.
- Add distilled water gradually.
- Mix thoroughly until the soil becomes a thick, creamy paste with water content above liquid limit.

2. **Filling the Shrinkage Dish**

- Grease the shrinkage dish lightly.
- Fill it with the soil paste in three layers, tapping gently to remove air bubbles.
- Level off the surface with a straight edge.

# *Procedure of Testing the Shrinkage limit*

**3. Drying the Soil**

- Allow the filled dish to air dry until the soil separates from the walls.
- Transfer to an oven at 105–110°C and dry for 24 hours.
- Cool in a desiccator.

**A. Measurement of initial Water Contents**

- Find the wet soil weight ($W_1$).
- After drying, measure dry soil weight ($W_2$).
- Compute water content:

$$w = \frac{W_1 - W_2}{W_2} \times 100$$

# *Procedure of Testing the Shrinkage limit*

**B. Volume of Wet Soil**

- Two methods available:
- (i) Mercury Displacement Method (most common)
- Fill a large tray with mercury.
- Press the shrinkage dish into mercury, level it, and collect displaced mercury.
- Weigh displaced mercury → convert weight to volume using density of mercury.





Shrinkage after oven dry











# *Procedure of Testing the Shrinkage limit*

**(ii) Wax Coating / Paraffin Method**

- Coat dry soil pat with melted wax.
- Measure coated volume by water displacement.
- Subtract wax volume to get soil volume.

**Volume of Dry Soil**

Repeat mercury displacement or wax method for the dry soil pat/ cake

**Formula for Shrinkage Limit (SL)**

$V_1$, $V_2$ are volume of wet and dry soil

W is the dry soil weight

$$\text{Shrinkage Limit (SL)} = w - \frac{V_1 - V_2}{W_2} \times 100$$

Volume of wet soil (V1)

| Test No. | 1 |
|---|---|
| Container id | S1 |
| Mass of empty shrinkage dish (g) | 41.54 |
| Mass of shrinkage dish + mercury (g) | 357.80 |
| Mass of mercury in shrinkage dish (g) | 316.26 |
| Density of mercury (g/cc) | 13.6 |
| Volume of shrinkage dish (cc) | 23.3 |
| Volume of wet soil (V1) (cc) | 23.3 |

Volume of dry soil pat (V2)

| Test No. | 1 |
|---|---|
| Container id | M |
| Mass of container + mercury (g) (before) | 741.1 |
| Mass of container + mercury (g) (after) | 539.1 |
| Mass of mercury displaced by dry soil pat (g) | 202 |
| Density of mercury (g/cc) | 13.6 |
| Volume of dry soil pat (V2) (cc) | 14.9 |

# Shrinkage Limit Test

| Test No. | 1 | 2 | 3 |
|---|---|---|---|
| Container id | S1 | S2 | S3 |
| Mass of container (m1) (in g) | 41.54 | 32.82 | 30.08 |
| Mass of container + wet soil (m2) (in g) | 81.47 | 71.67 | 69.15 |
| Mass of container + dry soil (m3) (in g) | 69.46 | 59.88 | 57.76 |
| Mass of dry soil (Ms = m3-m1) (in g) | 27.92 | 27.06 | 27.68 |
| Mass of water (Mw = m2-m3) (in g) | 12.01 | 11.79 | 11.39 |
| Moisture content (W) = (Mw/Ms*100) (%) | 43.0 | 43.6 | 41.1 |
| Volume of wet soil (V1) | 23.3 | 22.2 | 22.2 |
| Volume of dry soil pat (V2) | 14.9 | 14.1 | 14.4 |
| Shrinkage limit (%) | 12.9 | 13.7 | 13.0 |
| **Shrinkage limit (%)** | | | |

Shrinkage limit = W - (V1-V2)/Ms*100

